Sayed Ahmad Edriss Hossini

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RESEARCH PROFILE

Master's candidate in Software and Information Science (Social Systems Design) at Iwate Prefectural University, Japan. I design and build data-driven decision support systems that help institutions understand the people they serve and act on the evidence.

My Master's research created ISSDSS, a web-based decision support system that tracks international student satisfaction over time. It links survey analysis, AI-assisted analysis of open comments and a closed feedback loop, in which staff log interventions and the next survey round measures their effect. I presented this work at ICISS 2026 (University of Cambridge) and will present it at APIEMS 2026 in Busan, Korea.

I also bring one year of hands-on engineering at Takizawa Robotics, where I developed two generations of a crawler inspection robot, from the Arduino-based Crawler G1 to Crawler G2, a ROS 2 system with a React web interface delivered to a customer. For my PhD, I aim to bring explainable and causal AI to decision support systems and to physical AI systems such as robots.

EDUCATION

Master of Software and Information Science (Social Systems Design), Iwate Prefectural University, Japan — Apr 2025 – Mar 2027 (exp.)

- Thesis: *Decision Support System for International Student Satisfaction Management: A Longitudinal Case Study in Iwate Prefecture*
- Supervisor: Prof. Toshifumi Uetake · Co-supervisors: Prof. Kenichi Minamino, Prof. Takeo Takeno
- GPA: 3.0 / 4.0

Japanese Language Program, Morioka Information Business & Design College, Japan — Oct 2023 – Mar 2025

Bachelor of Management Information Systems, Faculty of Economics, Kabul University, Afghanistan — Mar 2017 – Dec 2021

RESEARCH INTERESTS

- Information Systems and Decision Support Systems
- Artificial Intelligence, Explainable AI and Causal AI
- Physical AI, Robotics and Autonomous Systems

PUBLICATIONS AND CONFERENCE PRESENTATIONS

Journal article

1. Abdali, S. M. Z., Sedeqi, A. R., **Hossini, S. A. E.**, & Qiami, M. H. (2023). The role of corporate social responsibility in hospitals: A study of Afghanistan's public and private hospitals. *The Journal of Social Science*, 7(13). doi:10.30520/tjsosci.1199829

Conference papers and presentations

2. **Hossini, S. A. E.**, & Uetake, T. (2026). Decision support system for international student satisfaction management: A longitudinal case study in Iwate Prefecture. *26th Asia Pacific Industrial Engineering and Management Systems Conference (APIEMS 2026)*, Busan, Korea, 29 Nov – 2 Dec 2026. Accepted for oral presentation (planned).
3. **Hossini, S. A. E.**, & Uetake, T. (2026). Exploring satisfaction factors of international students in Japan: A web-based decision support system, a longitudinal case study in Iwate Prefecture. *9th International Conference on Information Science and Systems (ICISS 2026)*, University of Cambridge, UK, 27–29 Aug 2026. Oral presentation.

4. **Hossini, S. A. E.**, & Uetake, T. (2025). A data-driven analysis of international student satisfaction through an information system: A case study in regional Japan. *33rd National Conference of the Japan Information-Culturology Society*, Japan, November 2025. Oral presentation.

RESEARCH EXPERIENCE

Master's Researcher, Graduate School of Software and Information Science, Iwate Prefectural University, Japan — Apr 2025 – present

ISSDSS: a web-based decision support system for international student satisfaction

- Designed and ran a two-round longitudinal survey in 2025–2026 (a pilot round, then the full survey in June 2026) at Morioka Japanese Language School and Iwate Prefectural University, measuring academic, daily-life, arrival-support and overall satisfaction.
- Analysed the data with descriptive statistics, reliability analysis, multiple regression and paired t-tests in Python and SPSS to identify the main drivers of satisfaction.
- Built ISSDSS end to end: PostgreSQL database for multi-institution, multi-period data; multilingual React and TypeScript interface for students and staff; dashboards with trend, bar and radar charts; PDF reports.
- Added an LLM layer (Gemini) that clusters free-text comments into topics with representative quotes.
- Designed a closed feedback loop (see, diagnose, act, measure, report): staff log each intervention in an Actions tab, and the next survey round measures its effect.
- Preparing qualitative interviews on international students' information needs and information-seeking behaviour.

Instructor and Research Assistant, Sultan Mahmud Ghaznavi Institute of Higher Education, Afghanistan — Jan 2022 – Aug 2023

- Taught Management Information Systems and related subjects; supervised student coursework.
- Researched corporate social responsibility in public and private hospitals: data collection, analysis, literature review and writing, leading to a peer-reviewed journal article (2023).

ENGINEERING AND INDUSTRY EXPERIENCE

Part-Time Engineer, Takizawa Robotics LLC, Takizawa, Japan — Jun 2025 – present

- **Crawler G1: research prototype** (April – July 2026). Tracked robot for remote inspection of tunnels, pipes and underground facilities. Designed the wiring and wrote the software: a Python operator interface on the PC, a C++ backend on the Arduino for motor control, and a network camera streaming over HTTP/IP, all linked to the PC by optical fiber.
- **Crawler G2: full deployment, delivered to a customer** (from July 2026). Rebuilt the robot on the ROS 2 platform with a Raspberry Pi and LiDAR, operated through a React web interface over an optical-fiber link of up to 500 m, with live front-camera video and LED lighting. Also designing the winch that winds and unwinds the fiber cable.

IT Intern, Avinton Japan — Dec 2025

- Supported system development, testing, debugging and technical documentation in a Japanese IT development workflow.

Intern, Tax Planning and Calculation, Ministry of Finance, Afghanistan — Sep – Dec 2020

TALKS AND EXHIBITIONS

- **CEATEC 2026**, Makuhari Messe, Chiba, Japan. Exhibited and gave live demonstrations of the crawler inspection robot at the joint booth of Takizawa Robotics and Iwate Prefectural University (booth 4H003).
- **Agile PBL Festival** (Agile PBL 祭り), Tokyo, November 2025. Speaker representing Iwate Prefectural University: presented our Android muscle fatigue monitoring app after my team won first place in the university's PBL competition.

SELECTED PROJECTS

Muscle fatigue monitoring with a wearable EMG sensor (Android, Kotlin) · Team leader · code 2025 – 2026

- Led the project-based learning (PBL) team that built the app; won first place in the university's PBL competition and presented it at the Agile PBL Festival in November 2025.
- Android app that connects to a BITalino board over Bluetooth and decodes its raw data frames (100 Hz, CRC-checked) to stream surface EMG in real time.
- Extracts mean, RMS and mean absolute value from each recording to score muscle condition, with a body map, live waveform, history and weekly summary.
- Companion web dashboard using Web Bluetooth (React, TypeScript) · code

University information system (Java, MySQL) 2021

- Designed the database tables, system workflows and user management functions.

AWARDS AND SCHOLARSHIPS

- Rotary Yoneyama Memorial Scholarship (host club: Mizusawa Rotary Club), April 2026
- Iwate Prefectural University Scholarship, 2025
- First place (Best Idea and Best Group Work), university project-based learning (PBL) competition, 2025, as team leader of the muscle fatigue monitoring app; selected to represent Iwate Prefectural University at the Agile PBL Festival, Tokyo

SKILLS

- **Programming languages:** Python, C++, Java, Kotlin, TypeScript, SQL
- **Frameworks and tools:** React, ROS 2, Android SDK, PostgreSQL, MySQL, Pandas, NumPy, SPSS, LLM APIs (Gemini)
- **Hardware:** Arduino, Raspberry Pi, LiDAR, BITalino EMG sensors, IoT devices
- **Research and analysis:** survey and questionnaire design; descriptive and inferential statistics; regression, hypothesis testing and reliability analysis; mixed methods; academic writing
- **Systems design:** decision support systems, requirements analysis, database design, system modelling

LANGUAGES

Persian/Dari (native) · English (professional; TOEIC 720, September 2024) · Japanese (JLPT N3, December 2024)

REFERENCES

- **Prof. Toshifumi Uetake**, Faculty of Software and Information Science, Iwate Prefectural University, Japan. Master's supervisor. Tel: +81 19-694-2644 · uetake@iwate-pu.ac.jp
- **Mr. Hayato Tajiri**, Takizawa Robotics LLC, Japan. tajiri@takizawa-robotics.com